Automated Sleep Stage Classification for Closed-Loop Lucid Dream Induction via CNN-LSTM on Single-Channel EEG

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***Abstract*—We present a closed-loop brain–computer interface (BCI) pipeline intended for lucid-dream-induction research, built on sleep stage classification from a single EEG channel. A CNN-LSTM hybrid divides each 30 s epoch into ten 3 s windows, extracts local features with a shared convolutional network, and models temporal structure with a bidirectional LSTM. On the Sleep-EDF Expanded database (153 whole-night recordings, 177,411 epochs) with class-weighted training, the model reaches 76.9% accuracy (κ=0.69) under a random 80/20 epoch-level split—a within-cohort figure that can overstate generalisation to new subjects—and 77.2% accuracy (κ=0.67) under leave-one-subject-out (LOSO) cross-validation on a 20-subject subset. REM F_1 is 0.75 on the full-dataset split, while N1 remains difficult (F_1=0.47). Our κ is below values reported for recent published systems, whose evaluation protocols differ. A software trigger with safety interlocks was functionally tested end to end on an Android device; stimulation efficacy was not evaluated. We also report a five-night, single-subject pilot (no EEG recorded) of wrist-vibration timing, as an anecdotal motivation for EEG-timed delivery. Code and pretrained models are released as open source.**

***Index Terms*—sleep staging, electroencephalography, CNN-LSTM, brain–computer interface, lucid dreaming, closed-loop stimulation**

I. Introduction

SLEEP disorders are highly prevalent [1], yet automated sleep staging remains challenging because of the complexity of electroencephalogram (EEG) signals and large inter-subject variability. Accurate stage detection is fundamental to clinical diagnosis and increasingly relevant to brain–computer interface (BCI) applications, including closed-loop neurostimulation.

Beyond clinical use, detecting sleep stages enables research on cue-based lucid dream induction. Lucid dreaming—awareness that one is dreaming—has been studied for nightmare treatment [2] and motor practice [3]. Many cue-based induction protocols deliver stimuli on fixed schedules, which motivates closed-loop alternatives that condition stimulation on the detected sleep stage.

We address two coupled problems: (1) sleep stage classification from single-channel EEG suitable for real-time use, and (2) trigger logic with safety interlocks for haptic stimulation. Our contributions are:

1) A CNN-LSTM hybrid for single-channel EEG, evaluated on the full 153-recording dataset (random split, κ=0.69) and under LOSO on a 20-subject subset (κ=0.67). These results are below recent published systems, which use different protocols (Table IV).
2) A software pipeline from preprocessing through classification to a phone-vibration trigger with safety interlocks, functionally tested on real hardware.
3) A comparison of four architectures on a 20-subject subset (single run).
4) A transparent, anecdotal single-subject pilot of vibration timing, with its limitations stated, motivating EEG-timed delivery.

II. Related Work

A. Deep Learning for Sleep Staging

Sleep staging is traditionally performed by expert visual scoring following the Rechtschaffen & Kales [4] and AASM [12] rules. Early automation relied on hand-crafted features; deep-learning approaches followed, for example stacked sparse autoencoders on time–frequency features [5]. DeepSleepNet [6] combined CNNs with bidirectional LSTMs on raw single-channel EEG and reported 82.0% accuracy (κ=0.76) on Sleep-EDF. SeqSleepNet [7] frames staging as sequence-to-sequence classification and reported κ=0.815 on the multi-channel MASS database (200 subjects). AttnSleep [8] applies multi-head self-attention to single-channel EEG and reported κ=0.79 on Sleep-EDF-20.

Our model follows the CNN–BiLSTM template of [6] in simplified form: it classifies one epoch at a time, without inter-epoch sequence modelling, and is oriented toward real-time use. We make no claim to improve on these systems.

B. Lucid Dream Induction and Sleep-Onset Stimulation

LaBerge et al. [9] reported that pre-sleep galantamine increased lucid dreaming frequency in a double-blind crossover study. Targeted stimulation at sleep onset has been explored with the Dormio device [15], which detects sleep onset and delivers cues. Our contribution in this area is limited to an open-source software pipeline and an anecdotal pilot; we do not evaluate induction efficacy.

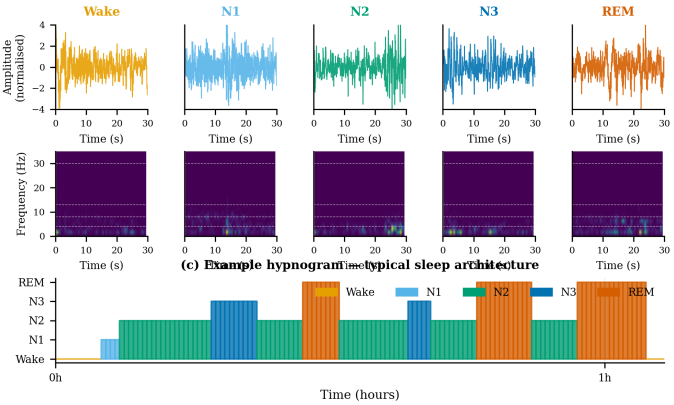


Fig. 6. EEG characteristics per sleep stage. Row 1: raw signal. Row 2: time-frequency spectrogram. Row 3: example hypnogram showing typical sleep cycle architecture.

Fig. 1. EEG characteristics per sleep stage. (a) Single-channel EEG (Fpz-Cz, 30 s). (b) Time–frequency spectrograms; delta (<4 Hz) is prominent in N3, theta (4–8 Hz) in REM, and sleep spindles (12–15 Hz) in N2. (c) Schematic hypnogram illustrating typical NREM–REM cycling.

TABLE I
SLEEP-EDF EXPANDED: EPOCH DISTRIBUTION (153 RECORDINGS)

Stage	Epochs	Percentage (%)
Wake	47,883	27.0
N1	21,522	12.1
N2	69,132	39.0
N3	13,039	7.3
REM	25,835	14.6
Total	177,411	100.0

III. METHODS

A. Dataset

We use the Sleep-EDF Expanded database [10], available from PhysioNet [11]. The experiments use 153 whole-night recordings (the Sleep Cassette subset, which contains 153 recordings from 78 healthy participants aged 25–101 years) and the single EEG channel Fpz-Cz. The final model's random-split evaluation uses all 153 recordings; the architecture comparison (Table II) and the LOSO validation use a 20-subject subset.

Preprocessing. Raw EEG was bandpass filtered at 0.3–35 Hz with a 4th-order Butterworth filter, consistent with the AASM recommended range [12], and segmented into 30 s non-overlapping epochs (3,000 samples at 100 Hz) aligned to hypnogram annotations. Stages were mapped to the 2007 AASM standard: Wake (W), N1, N2, N3 (N3 and N4 merged) and REM. Epochs before the first and after the last non-Wake annotation were excluded, and unknown-stage and movement-time epochs were discarded. Fig. 1 shows example EEG characteristics per stage.

Dataset statistics. Table I and Fig. 2 summarise the resulting epoch distribution.

B. CNN-LSTM Architecture

Fig. 3 illustrates the architecture. Each 30 s epoch is processed as follows.

Step 1 – Window segmentation. Each 3,000-sample epoch is divided into N_w=10 consecutive windows of 300 samples

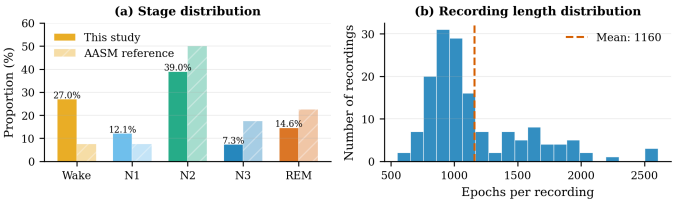


Fig. 2. Dataset statistics. Sleep-EDF dataset (5 classes, 177,411 epochs total).

Fig. 2. Dataset statistics. (a) Stage distribution compared with AASM reference ranges. (b) Recording-length distribution across the 153 recordings (mean: 1,159 epochs per recording).

Fig. 1. CNN-LSTM hybrid architecture for sleep stage classification. Each 30-second EEG epoch is divided into ten 3-second windows processed independently by a shared CNN before temporal modelling with bidirectional LSTM.

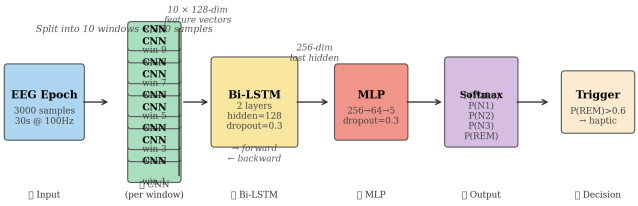


Fig. 3. CNN-LSTM hybrid architecture. Each 30 s epoch is divided into ten 3 s windows processed by a shared CNN before temporal modelling with a bidirectional LSTM.

(3 s), a length chosen to capture individual sleep oscillations: spindles (0.5–2 s), K-complexes (∼0.5 s) and slow waves (∼1 s).

Step 2 – CNN feature extraction. A shared three-block 1D CNN processes each window independently:

$$\mathbf{f}_t = \mathrm{CNN}(\mathbf{x}_t), \quad t = 1, \ldots, N_w, \tag{1}$$

where $\mathbf{x}_t \in \mathbb{R}^{300}$ and $\mathbf{f}_t \in \mathbb{R}^{128}$. Each block comprises Conv1D, BatchNorm, ReLU and MaxPool layers with 32, 64 and 128 filters.

Step 3 – Temporal modelling. The sequence $\{\mathbf{f}_1, \ldots, \mathbf{f}_{N_w}\}$ is fed to a two-layer bidirectional LSTM with hidden dimension 128:

$$\mathbf{h}_t = \mathrm{BiLSTM}(\mathbf{f}_t, \mathbf{h}_{t-1}). \tag{2}$$

The final hidden state $\mathbf{h}_{N_w} \in \mathbb{R}^{256}$ summarises the epoch.

Step 4 – Classification. A two-layer MLP (256→64→5) with softmax produces class probabilities.

C. Training

Models were trained with Adam [13] (learning rate 10^{-3}, weight decay 10^{-4}) for 60 epochs with batch size 64. Balanced class weights were applied to the cross-entropy loss. A ReduceLROnPlateau scheduler (patience 5, factor 0.5) and gradient clipping at norm 1.0 were used.

D. Closed-Loop Trigger System

The deployed software chain has three components.

Detection. Real-time inference produces $P(\mathrm{REM})$ for each incoming epoch, smoothed over a 3-epoch sliding window, with threshold $P(\mathrm{REM})>0.6$.

Decision logic. A multi-condition safety interlock gates stimulation: (i) $P(\mathrm{REM})>0.6$ sustained for $\geq$30 s, (ii) no

TABLE II
ARCHITECTURE COMPARISON (20-SUBJECT SUBSET, RANDOM SPLIT, SINGLE RUN)

Exp.	Architecture	Acc.	κ
001	SleepLSTM	0.7685	–
002	CNN spectrograms	0.7167	–
003	CNN-LSTM (proposed)	0.8014	0.7128
004	CNN+Transformer	0.7900	0.7003

TABLE III
PER-CLASS RESULTS, CNN-LSTM, 153 RECORDINGS, RANDOM 80/20 EPOCH SPLIT (TEST EPOCHS: 35,483)

Stage	Prec.	Rec.	F_1	Support
Wake	0.95	0.85	0.90	9,577
N1	0.43	0.51	0.47	4,304
N2	0.87	0.76	0.81	13,827
N3	0.60	0.91	0.73	2,608
REM	0.71	0.80	0.75	5,167
Macro	0.71	0.76	0.73	
Weighted	0.79	0.77	0.78	

Wake epochs in the preceding 60 s, and (iii) a single stimulation per sleep cycle (300 s cooldown). A variant gating on stable N1 rather than REM is planned (Sec. IV-E).

Output. Upon trigger, haptic feedback is delivered to an Android device via the JOIN notification API [14].

E. Evaluation Protocol

Random split. An 80/20 stratified split of pooled epochs with fixed seed (42). Because the split is at the epoch level, epochs from the same recording can appear in both training and test sets; this figure therefore represents within-cohort performance and is optimistic with respect to new subjects.

LOSO. Leave-one-subject-out cross-validation on a 20-subject subset, which estimates generalisation to unseen individuals.

IV. EXPERIMENTS

A. Architecture Comparison

Table II compares four architectures on a 20-subject subset (single run, random split). The CNN-LSTM hybrid obtained the highest accuracy, but the gap to the CNN+Transformer model is small (0.8014 vs. 0.7900) and comes from a single run, so we do not claim a reliable ordering between the top two. The spectrogram-CNN model scored lowest in our setup.

B. Detailed Results

Table III reports per-class results for the CNN-LSTM on the 153-recording random split, computed from the confusion matrix in Fig. 4 (accuracy 0.7689, Cohen's κ=0.6932). N3 has the highest recall (0.91) and N1 the lowest (0.51), consistent with its brief duration and spectral overlap with Wake and N2. Per-class ROC curves are shown in Fig. 5; the macro-average AUC is 0.95, with N3 highest (0.985) and N1 lowest (0.860).

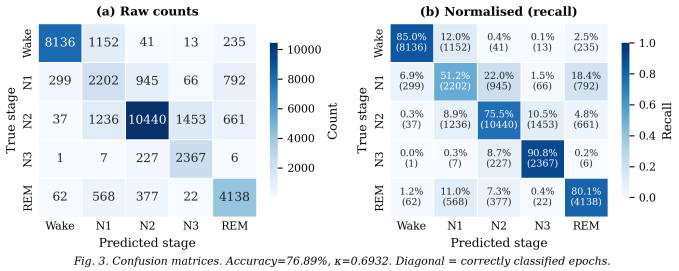


Fig. 3. Confusion matrices. Accuracy=76.89%, κ=0.6932. Diagonal = correctly classified epochs.

Fig. 4. Confusion matrices (153 recordings, random split). (a) Raw epoch counts. (b) Normalised by true class (recall).

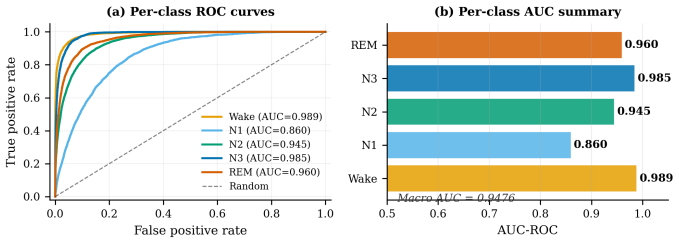


Fig. 4. Receiver operating characteristic (ROC) curves and AUC per sleep stage. One-vs-rest classification.

Fig. 5. One-vs-rest ROC curves and per-class AUC for the CNN-LSTM. N3 achieves the highest AUC (0.985) and N1 the lowest (0.860).

TABLE IV
COMPARISON WITH PUBLISHED SINGLE-CHANNEL RESULTS. LITERATURE VALUES ARE AS REPORTED BY THE ORIGINAL AUTHORS; PROTOCOLS ARE NOT IDENTICAL.

Method	Dataset / protocol	Ch.	κ
Ours, LOSO	Sleep-EDF, 20 subj.	1	0.67
Ours, random split	Sleep-EDF, 153 rec.	1	0.69
DeepSleepNet [6]	Sleep-EDF	1	0.76
AttnSleep [8]	Sleep-EDF-20	1	0.79

C. Comparison with Published Results

Table IV places our results beside values as reported by the original authors. Our LOSO κ (0.67) is lower than both published single-channel results. Datasets, subject counts and evaluation protocols differ, so the table indicates where this baseline stands rather than offering a like-for-like ranking.

D. Closed-Loop Functional Test

The trigger chain was tested end to end: when the safety conditions were satisfied, the system delivered haptic feedback to an Android device. This establishes that the software chain executes; it does not evaluate real-time classification accuracy on a live sleeper or any effect on dreaming.

E. Pilot Self-Experiment

As a preliminary, non-validating observation, one subject (the author) received single-pattern phone vibration at the wrist on five nights. Stimulus timing was set by elapsed time after lying down; no EEG was recorded, so stage labels are *estimated*. Perception was self-reported on waking as *blended* (felt integrated into the body) or *device* (felt like external hardware).

TABLE V
PILOT SELF-EXPERIMENT: STIMULUS TIMING VS. PERCEIVED MODALITY (LABELS ESTIMATED FROM ELAPSED TIME; NO EEG RECORDED)

Night	Estimated stage	Perception
1	≈N1	Blended
2	≈Wake	Device
3	≈N1	Blended
4	≈N1	Device
5	≈N1	Device

Two of four nights with estimated-N1 timing were reported as blended; the single estimated-Wake night was reported as device (Table V). With n=5, one unblinded subject and subjective reports, no statistical inference is possible. These observations motivate EEG-timed delivery in future work, consistent with targeted sleep-onset stimulation studies [15]. A detailed dream with specific visual content (a clock reading 14:15) was also recalled on one night during the experimental period (the night was not recorded); we report this only as an anecdote. The pilot targeted sleep onset (N1), whereas the trigger in Sec. III-D conditions on REM; an N1-targeted trigger is future work.

V. DISCUSSION

A. Architecture Comparison

The CNN-LSTM obtained the highest accuracy in our comparison, but the margin over the Transformer variant is small and from one run. A possible explanation is that the LSTM's sequential inductive bias helps at modest dataset sizes; we did not test this.

B. N1 Classification

N1 is the hardest stage (12.1% of epochs in Table I; F_1=0.47 on the 153-recording split). Its EEG overlaps with Wake (residual alpha activity) and N2 (emerging theta). On the 20-subject subset, the hybrid's N1 F_1 was 0.47 compared with 0.35 for the LSTM-only baseline (a +34% relative difference, single run), which may indicate that local frequency features from 3 s windows help for N1.

C. Implications for Deployment

Under LOSO our κ is below recent literature (Table IV). Single-channel operation is attractive for wearables, but consumer headbands use different montages from Fpz-Cz, so transfer would require adaptation and validation on new recordings. Weak N1 detection (F_1=0.47) is also relevant to the planned N1-targeted trigger, which would likely need subject-specific calibration.

D. Limitations

(i) Evaluation is offline; real-time validation on a sleeping person with a wearable has not been performed. (ii) The random-split figures are epoch-level and therefore optimistic about unseen subjects; LOSO was run only on a 20-subject subset. (iii) The cohort is healthy and spans ages 25–101; performance in other populations is unknown. (iv) Haptic stimulation efficacy for lucid-dream induction was not measured. (v) The self-experiment is an anecdotal, unblinded, single-subject observation without EEG and is not evidence of induction efficacy.

VI. CONCLUSION

We presented a CNN-LSTM sleep-stage classifier for single-channel EEG and a software trigger pipeline with safety interlocks. On 153 recordings it reaches κ=0.69 under an epoch-level random split and κ=0.67 under LOSO on a 20-subject subset, below recent published systems; N1 detection remains weak. The pipeline and models are released as open source as a baseline for closed-loop sleep research and form the first component of the LUCID: Reality? project. Future work includes real-time validation with a consumer EEG headband, EEG-timed N1 and REM triggers, and a controlled study of stimulation effects.

CODE AND DATA AVAILABILITY

Code, pretrained models and experiment logs: https://github.com/anu-rag-007/PROJECT-07. Preprint (Zenodo): https://doi.org/10.5281/zenodo.21885881. Dataset: PhysioNet Sleep-EDF Expanded, https://physionet.org/content/sleep-edfx/1.0.0/.

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